

Multi-source inputs

Evidence: DEP
stormwater polygons
(categories 1–2)

Evidence: NYC 311
street-flooding
complaints

Evidence: USGS Ida
high-water marks

H3 assembly (R9)

Join layers to H3 cells
+ static predictors

Labels: evidence-
positive / evidence-
unrecorded (not
verified flood)

Learning & validation

Gradient-boosting
classifier + evidence-
score regressor

H3-block GroupKFold
spatial CV
(R7 parent blocks)

$S_h(c) = f_{\theta}(X_c)$
susceptibility score
(rainfall deferred)

Diagnostics & outputs

Source ablation &
block-size / LOBO

Scale-loss Jaccard
(domain-masked R10)

Sandy coastal OOF
diagnostic

FEMA Sandy negative control
(never a training label)

